Introduction to API

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Agenda

In this session, we will discuss:

- Applications that Dominate the Market
- Introduction to API
- Components, Types, and Examples of APIs
- Working of API
- REST APIs
- Role of API in Application Development
- API Endpoints and Security

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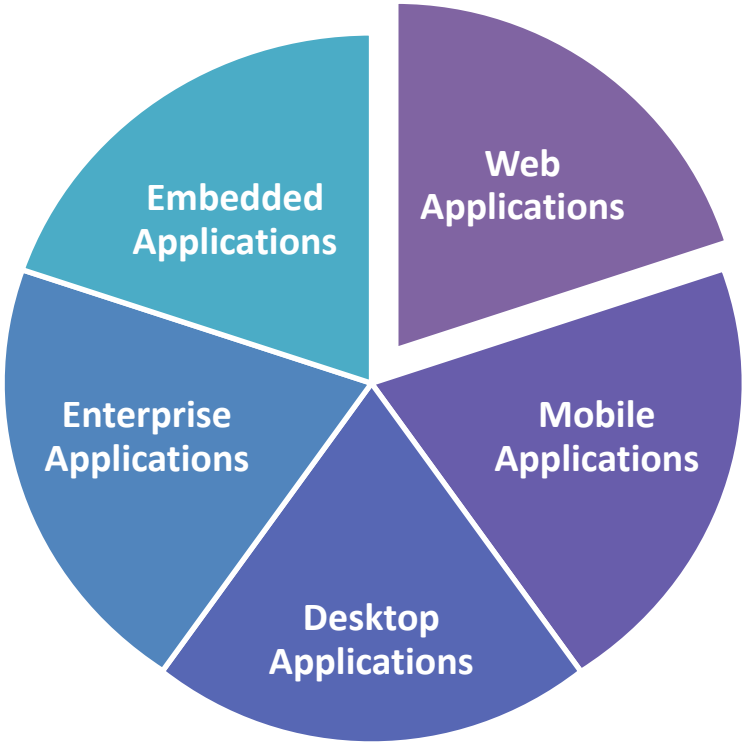
Applications that Dominate the Market

An application is a software program or piece of code designed to execute specified tasks or functions for the end users.

Types of Applications



AutoCAD

Dropbox

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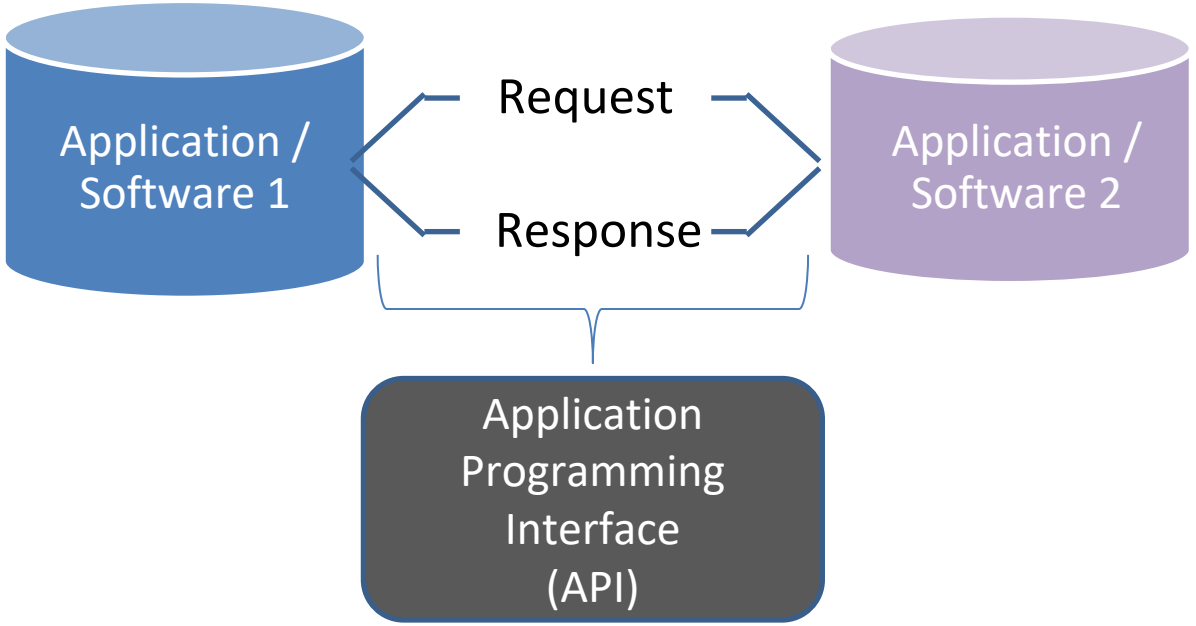
Introduction to API

- **API = Application Programming Interface**
  - Application: Any software with a distinct function.
  - Interface: Contract of service between two applications.
- API serves as a bridge between different software applications, enabling seamless communication and interaction.
- For example: FB API, Logon API, etc. are used to fetch required data from the database.



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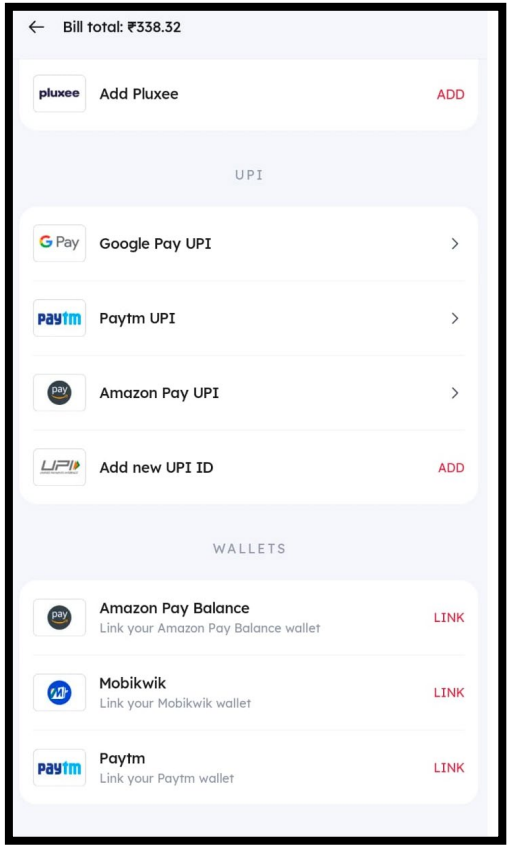
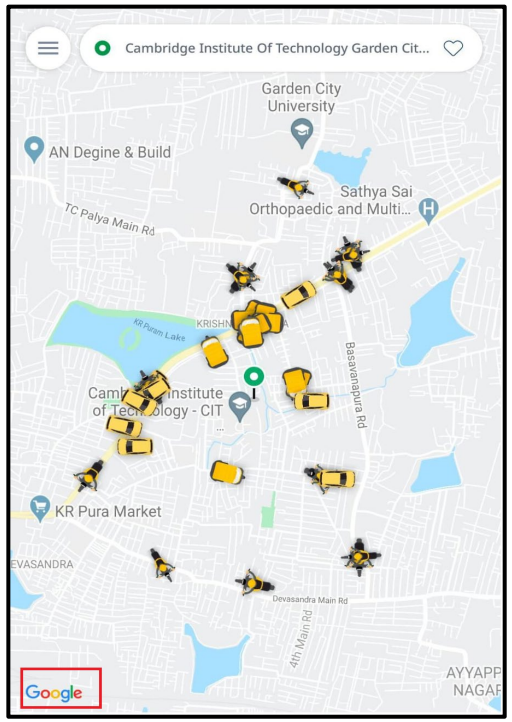
Examples of API

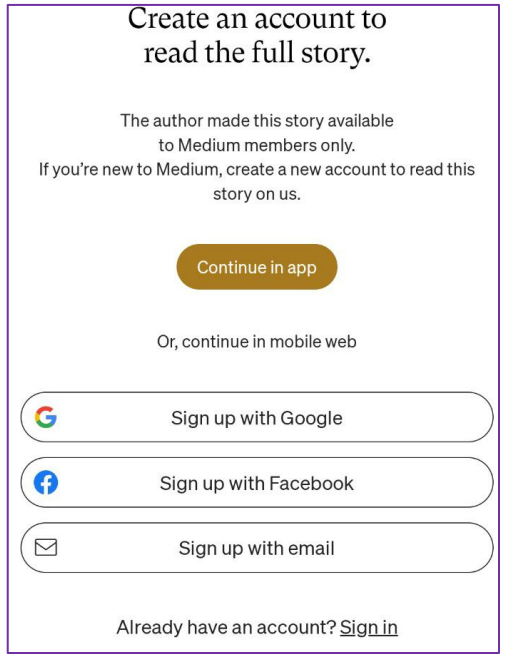
Payment API



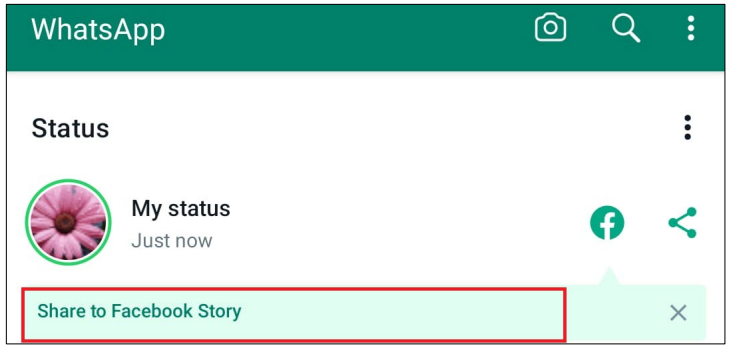
G Map API



Logon API



FB API



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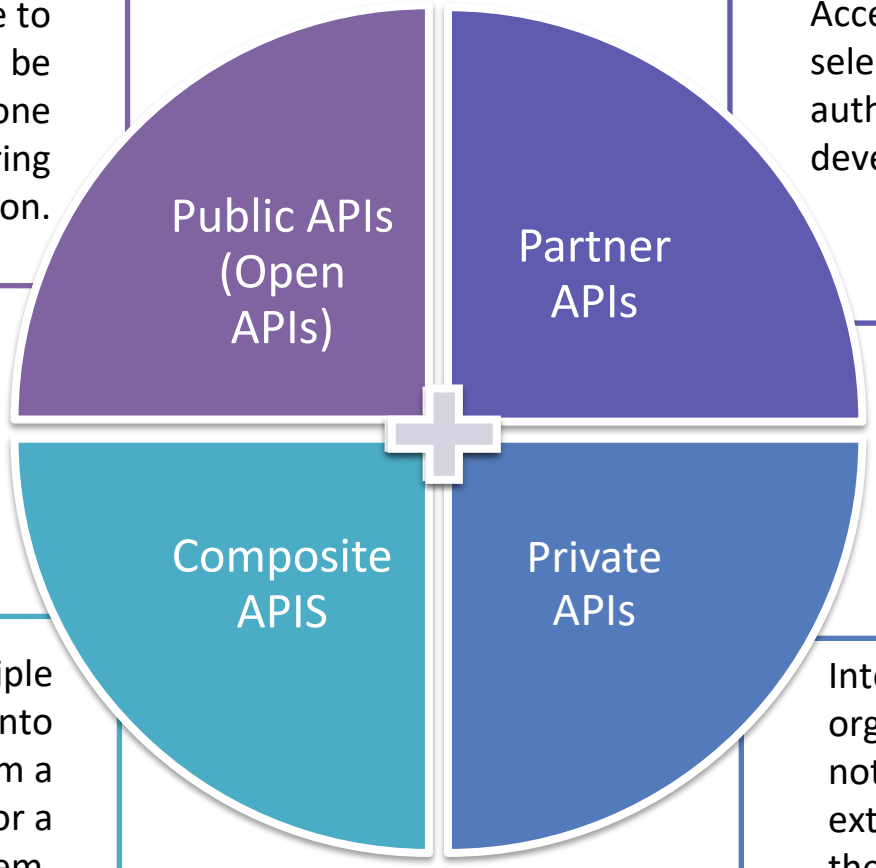
Types of API

Freely available to developers and can be accessed by anyone without requiring explicit permission.

Accessible only to selected partners or authorized third-party developers.



Combine multiple API requests into one call to form a solution for a complex problem.

Internal to an organization and are not made available to external developers or the public.

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Components of API

- Endpoint (**https://api.example.com/users**)
- Methods
  - ✓ GET
  - ✓ POST
  - ✓ PUT
  - ✓ PATCH

API Request

API Server

- Receiving Request
- Data Processing
- Security Enforcement
- Response Generation

- Status Code
  - ✓ 200
  - ✓ 400
  - ✓ 401
  - ✓ 404
  - ✓ 500
- Authentication (API Keys)

API Response

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Working of API

Client-Server Model

- Requester is called client
- Responder is called server

API Works in Four Different Ways

- SOAP APIs
  - Simple Object Access Protocol
  - Message exchange using XML
- RPC APIs
  - Remote Procedure Calls
- WebSocket APIs
  - Uses JSON objects to pass data
- REST APIs
  - Representational State Transfer Application Programming Interface

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REST APIs

Representational State Transfer

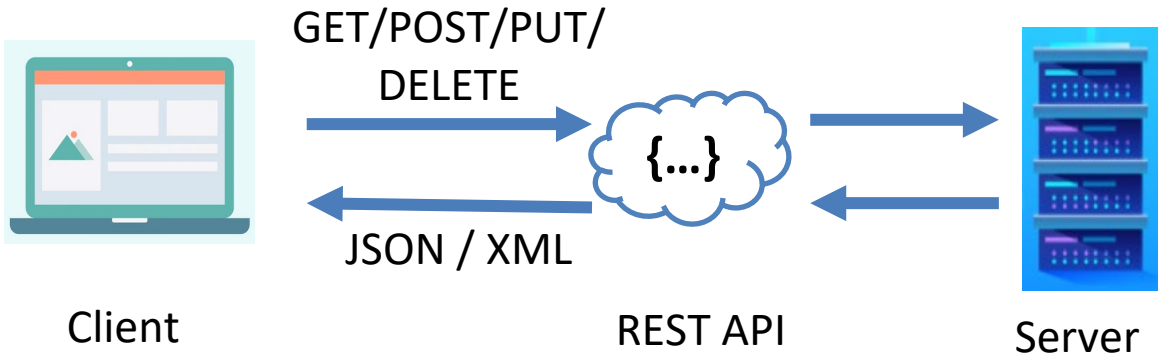
Client can use GET, POST, PUT, DELETE functions to access server data

Data is exchanged using HTTP

Main feature – Statelessness; which means client data will not be saved on server in between requests

Client request to server is similar to typing URL in browser

Server response is in the form of plain data



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Role of API in Application Development

API Advantages

- Integration of new applications with existing software
- Rapid deployment of innovative services
- Easy expansion across platforms
- Easy to maintain

API Types

- Private APIs – used within an enterprise
- Public APIs – open to public
- Partner APIs – available to authorized external developers/enterprises
- Composite APIs – combination on 2 or more APIs to solve a complex business case

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API Endpoints and Security

API Endpoints

- Endpoints are touchpoints in API communication.
  - They are important from security perspective as they make systems vulnerable to attacks.
  - They can also impact performance by becoming the bottlenecks in case of high traffic.

How to secure an API?

- **Authentication tokens** can be used to authorize users to make API calls. For example, email login.
- **API keys** can be used to verify the application or program making API call.

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Example: Weather Map API

API Request/Call:

https://api.openweathermap.org/data/2.5/weather?q=London&appid=YOUR_API_KEY

- Protocol – HTTPS
  - Ensures secure communication between the client application and the API server.
- Domain Name – api.openweathermap.org
  - Identifies the server hosting the OpenWeatherMap API, which provides weather data.
- Path – (/data/2.5/weather)
  - This specifies the location of the weather data resource within the API.
- Query String – ?q=London&appid=YOUR_API_KEY
  - This provides additional parameters to refine the request:
- appid=YOUR_API_KEY: This parameter is crucial for authentication.
  - The API server uses this key to identify your application and grant access to the requested data.

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Summary

Here is a brief recap:

- APIs act as messengers between applications, enabling them to share data and functions.
- Developers use APIs to access pre-built features like maps or payments, saving time and effort.
- The different types of APIs are public, private, partner, and composite APIs.
- From weather to social media, APIs unlock data for countless applications.
- Tech giants like Google and Amazon offer public APIs for developers to build new things.
- Modern software development relies on APIs as building blocks to create features.

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